

Review

A Brief History of Milk Hygiene and Its Impact on Infant Mortality from 1875 to 1925 and Implications for Today: A Review

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ABSTRACT

The objective of this review is to provide an integrated historical account of the complex, often convoluted events impacting milk hygiene and its resultant effect on infant mortality from 1875 to 1925. Heat pasteurization of cow's milk is necessary for rendering this important nutrient source safe for humans—particularly infants. Developed by Louis Pasteur in 1864, pasteurization evolved from the commercially important parboiling of wine and beer when the Industrial Revolution was effecting rapid societal change in Western societies. In European and American societies of the early and mid-19th century, infant mortality rates were 30- to 60-fold higher than the current rates of five or six deaths per 1,000 live births per year. With proof of the germ theory of disease came convincing evidence of the role of microbes in the transmission of infections, which led to the discovery that microbial pathogens were transmissible via milk. Diseases caused by milkborne pathogens include human and bovine tuberculosis, brucellosis, salmonellosis, streptococcal infections, diphtheria, and “summer diarrhea.” With pasteurization of milk, infectious diseases with their high infant mortality rates decreased by only half by the early 20th century, despite concurrent medical and dairy hygiene advances. To further mitigate unacceptably high infant mortality rates, social support providers—including public health nurses and midwives—encouraged breastfeeding, especially among socioeconomically disadvantaged mothers. Improvements in pulsating vacuum milking machines also favorably impacted food safety by providing a clean, enclosed environment. Currently, bottle feeding still competes with breastfeeding as the preferred method, and the sale of raw, unpasteurized milk remains a contentious issue. Informed and responsible food safety professionals, physicians, and public health officials currently view breastfeeding as the preferred feeding method and milk pasteurization as the safer and more prudent alternative.

Key words: Infant mortality; Milkborne diseases; Pasteurization

IMPACT OF THE INDUSTRIAL REVOLUTION ON MILK HYGIENE AND FAMILY STRUCTURE

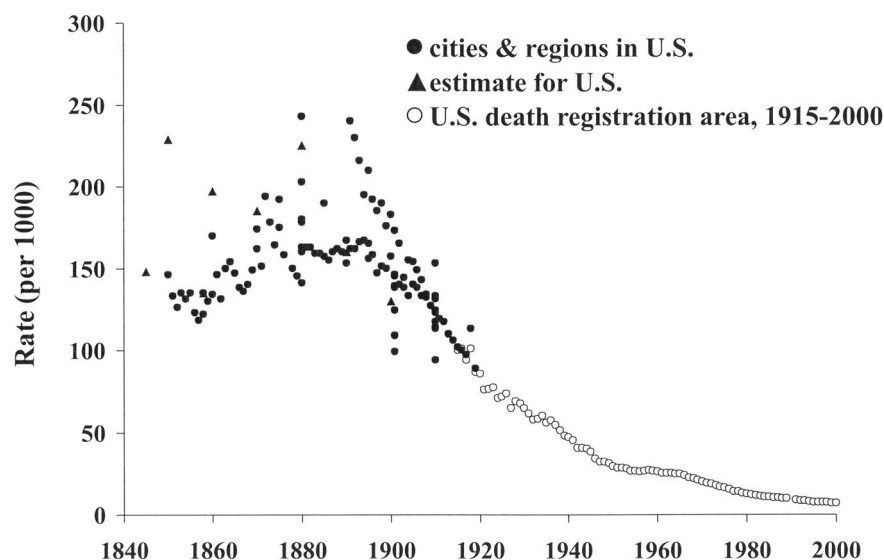
During the middle decades of the 19th century, the rapidly industrializing European nations and the United States experienced increasing rates of infant mortality (Fig. 1) (3, 35, 40, 43). Early during this period, European societies had high infant mortality rates of 150 to 300 deaths per 1,000 live births per year, compared with 5 or 6 deaths annually today, with lower rates in rural areas. Important social influences at the time included rapidly expanding industrialization with resultant urban relocation of rural populations. In the United States, these changes occurred at a time of accelerating foreign immigration that further increased urbanization. During this period, the transmission of infectious diseases was facilitated by increased urban crowding in substandard housing, poor environmental sanitation, changes in the source and processing of commercial milk, the introduction of cow's milk formulae,

and lack of scientific information about the causes of infectious disease.

From 1840 to 1860, several factors were primarily responsible for the decline that occurred in the wholesomeness of cow's milk, including the dairy industry's expansion during urbanization as brought about by the Industrial Revolution. This expansion was accompanied by a departure from traditional small dairy herds housed relatively close to consumers, often in open areas of cities, e.g., the Boston Common with its limit of 70 cows. The new dairy herds of the Industrial Revolution were large, with as many as 2,000 cows confined in cramped urban quarters. A major economic factor in this transition was that these larger dairy herds were exclusively fed “slop house” distillery waste in the notorious “swill dairies” (see below). At this time, other contributors to the decline in the wholesomeness and safety of cow's milk were inadequate refrigeration, the absence of milk processing standards, and fraudulent practices such as mixing in additives to allegedly “salvage” or “enhance” the increasingly poor quality of milk available to infants and families. Despite the evolution of these practices, it was not until the 1870s when cow's milk was first linked

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FIGURE 1. U.S. infant mortality rates, 1840 to 2000 (reproduced with permission (35)).



*Data source: S. H. Preston and M. R. Haines (1991); Alter (1997); U.S. Census office (1902-2000)

epidemiologically to typhoid fever and other infectious conditions (see below). At the same time, a concomitant decline in breastfeeding in many countries was observed despite prior recognition in Norway and Sweden a half century earlier of the association between increases in infant mortality and increases in the use of cow's milk for infant feeding (44).

As the Industrial Revolution progressed into the mid-19th century, the practice of feeding distillery mash (slop) to dairy cattle greatly expanded. This led to a deterioration in the quality of cow's milk. Frederick Stenn (61), in his eloquent address in 1978 to the Chicago Literary Society, offered the following vivid description of the distilleries of the beer, wine, and spirits industry of the period and their swill milk by-product sold commercially for additional profitability: "City milk was almost entirely distillery milk. These distilleries housed up to 2,000 head of cows, such as prevailed in Johnson's Grain Distilleries of Manhattan. Cow stalls rented for five dollars a year, the cost of mash, two dollars a month, labor paid in whiskey. The slop—slop it certainly resembled—was conveyed from large elevated tanks into wooden sluices leading to the cow stalls 3 feet wide. The cow consumed 32 gallons of slop and 3 pounds of hay. Ulcers developed in the mouths of the cows; their tails fell off. Tuberculosis of the glands, lungs, and intestines followed. One Brooklyn distillery indicated that, out of 1,811 cows, 230 died in 10 weeks. Milk so obtained was pale blue, often turbid, and malodorous. Peculating dairymen concealed their wickedness, not so much as by diluting the milk with water, as by adulterating the swill milk with plaster of Paris, charcoal, starch, sugar, flour, and egg—making bad matters worse. No wonder the Olympus of pediatrics, Abraham Jacobi, and his colleagues advised mothers to 'give no milk.' "

Changes in the family structure in the mid-19th century—another consequence of industrialization with its economic forces—also contributed to the decline of breastfeeding, perhaps by as much as 50 to 70%, as mothers

were increasingly forced to work outside the home. In response, commercial cow's milk-based infant formulas were introduced. These formulas evolved into the mainstay infant feeding practice in the latter part of the 19th century, a trend that has continued to the present (3, 49). In the final analysis, cows represent foster mothers of the human race or more appropriately wet nurses for infants and children.

DISCOVERY AND INCREASING AWARENESS OF THE ROLE OF MICROORGANISMS IN MILKBORNE DISEASE

Fortuitously, between 1850 and 1875, a tumultuous period of industrialization, several important developments evolved that contributed to a rapid decline in the alcohol industry's swill dairies in both Europe and North America. Although the rise of the temperance movement was one factor in this decline, a more important factor was the work that took place in the separate laboratories of two brilliant scientists: Louis Pasteur in France and Robert Koch in Germany. Late in the 1860s, while demonstrating the role of microbes in the fermentation of beer and wine, these two scientists provided convincing evidence for the basis of the germ theory of disease. As a consequence of their work, the influence of the anticontagionists, who believed that illness resulted from adverse environmental influences referred to as "miasmas" or from contaminated "bad air" containing particles of decaying organic matter, was greatly diminished.

In their separate research, Pasteur and Koch both conclusively demonstrated the causative role of microorganisms, not miasmas, in the transmission of infectious disease. Specifically, the experiments of Pasteur in 1865 on silkworm disease and those of Koch in 1876 on anthrax firmly established the role of microorganisms in these diseases (33, 34). These scientists forever changed the worlds of medicine and science; they opened the door to new ways of thinking about how to significantly reduce the morbidity and mortality associated with milkborne and person-to-person disease transmission.

Pasteur's work from 1860 to 1864 was focused on the wine and beer industries, not on milk hygiene. He demonstrated that heating wine and beer to 140 to 212°F (60 to 100°C) protected these beverages from spoilage. In 1870, pasteurization was employed in the United States by the Anheuser-Busch Brewery. This process permitted rail car transport of beer from St Louis to Texas without spoilage.

IMPLEMENTATION OF MILK PASTEURIZATION: PROBLEMS AND SOLUTIONS

In 1870, Professor N. J. Fjord in Denmark was the first person to adapt the process of pasteurization for milk. It was in Germany, however, in 1882 that the first commercial pasteurization process was developed (32, 59). These milestones revolutionized the processing, transport, and storage of dairy products. By 1885, heat treatment of market milk was also being applied by the Swedes and Danes to preserve and assure the safety of milk using this new process, named pasteurization after its discoverer. Immediately thereafter in 1886, Franz von Soxhlet, a distinguished German chemist, was the first to design and fabricate equipment for the heat treatment of bottled milk (the Soxhlet apparatus), which led to popularization of the process and application in the home (40, 44, 53). In 1889, Abraham Jacobi, a German physician working in New York City (NYC) and considered the father of American pediatrics, became an adamant and vocal supporter of the virtues of pasteurizing milk in the home and endorsed the use of the Soxhlet apparatus with sealed nursing bottles (29).

Despite the benefits of pasteurization, most dairies were initially reluctant to adopt practices requiring additional investment in equipment, time, and effort. Nonetheless, between 1895 and 1910 the public health advantages of pasteurization became increasingly obvious. This awareness prompted well-capitalized larger dairies in larger urban settings in the United States, Canada, and the United Kingdom to seek a competitive advantage by investing in pasteurization processes and equipment before smaller independent dairies elsewhere could do so (53, 69). By 1911, the cities of Chicago (31) and New York mandated pasteurization of cow's milk.

Proponents of pasteurization became increasingly influential during the 1890s. Chief among these was New York businessman Nathan Straus and his brothers, the proprietors of Macy's Department Store in NYC. At the time, Nathan Straus was the most visible and vocal advocate for pasteurization in the United States and Europe. He and his wife, Lena, were motivated by the loss of two of their children to milkborne tuberculosis (23). Nathan Straus traveled widely to speak about the need for pasteurization to protect infants and children against preventable infectious diseases. Straus also subsidized milk pasteurization for disadvantaged children living in New York tenement districts. In *Disease in Milk: The Remedy Pasteurization: The Life Work of Nathan Straus*, Lena Straus summarized her husband's efforts promoting pasteurization (62).

Despite pasteurization's public health advantages, many physicians opposed it, believing that pasteurization dimin-

ished the nutritional value of milk (40). As a result, during the final decade of the 19th century when it first became possible to analyze the chemical composition of milk, percentage feeding (based on percentages of fats, carbohydrates, etc., in milk) came into vogue. Physicians developed elaborate schemes to allegedly augment the nutritional composition of cow's milk for human infants, which frequently included utilizing arcane proprietary formulae. As a result, the term "baby-feeding fluids" was changed to "formula," a designation still used today.

The movement to augment cow's milk for improved nutritional composition began in Germany with Justus von Liebig, who demonstrated that foods are composed of various percentages of fats, carbohydrates, and proteins, with each meeting specific nutritional needs of infants and persons in other age groups (40). These and other analytical developments in food science provided a basis for devising food equivalents. In 1893, Harvard professor Thomas Rotch declared that "the preventive medicine of early life is preeminently the intelligent management of nutriment" (40). Rotch developed elaborate formulae to modify cow's milk to closely resemble human milk. Among many opponents of applying the algebraic formulae of Rotch and others in the preparation of infant feedings was Abraham Jacobi, who contradicted this approach in his lectures: "You cannot feed babies with mathematics; you must feed them with brains" (28).

To complicate the implementation of milk pasteurization, early pasteurization processes did not always yield the desired outcome. The flash process utilized a processing temperature that was too high, thereby rendering milk with a "cooked" taste. This process could also result in incomplete pasteurization due to incomplete mixing, thereby permitting some pathogenic bacteria to survive. In 1907, an understanding of these problems eventually prompted dairies to convert to vat or batch pasteurization processes, which then became the accepted standard (52, 53) (Table 1).

Most turn-of-the-century physicians remained skeptical regarding the merits of pasteurization. Instead, they wanted to see the development of ultrahigh standards for dairy operations (40). These standards included assuring the purity of raw milk from clean cows, employing illness-free handlers, and using clean and appropriate utensils. Physicians argued that only raw milk derived under these improved circumstances should be incorporated into infant formulae. In 1891, this approach was first formalized in Boston and became known as the Harvard plan.

In 1893, Henry L. Coit, an influential New Jersey pediatrician whose son had died from milkborne illness, realized that the milkborne disease problem could not be solved by legislation (68). Thus, he modified the Harvard plan and established the first Certified Milk dairy in the United States. In response, many physicians supported these certified products for treating sickly infants. Coit also established the American Association of Medical Milk Commissions (AAMMC), a national program staffed by volunteer physicians to certify that milk was produced under extremely high standards. Each operation required oversight by a committee of knowledgeable experts including veterinarians and "sanitarians," the latter known today as

TABLE 1. *Evolution of methods for milk pasteurization: 1880 to present*

Period	Predominant pasteurization method	Advantages	Disadvantages
1880–1895	Raw milk in individually sealed bottles immersed in boiling water or heated sand	Suitable for home use	Requires specialized equipment, e.g., Soxhlet apparatus; gives milk a “cooked” taste when taken to sterility (53)
1895–1910	Flash pasteurization on continuous basis: milk heated to 178°F (ca. 81°C) and then chilled at once	Process is easy to establish commercially	Inconsistent results; some bacterial pathogens survive
1910–1940	Batch method: milk heated to 140°F and held 20 min	Improved flavor; assurance all pathogens destroyed, including those causing tuberculosis	Time-consuming and cumbersome on a commercial basis
1940–present (North America)	High temperature–short time gradients: heated to 160 to 162°F (ca. 71 to 72°C) for not less than 15 s	Efficient and less time-consuming; extended 14-day shelf life	Flavor as good as batch method
1975–present (Europe)	Ultrahigh temperature: heated to 275°F (135°C) for 1–2 s; common in Europe, including ~95% of milk consumed in France, Belgium, Portugal, and Spain	Product can be stored without refrigeration for up to 6 mo	Does not taste as fresh; requires special sterile containers

environmental health officials. This oversight caused the retail price of milk to double because the standards specified were extremely expensive and thus difficult for most dairy operations to manage. Nonetheless, the Certified Milk dairy program was successful in providing leadership needed to effect improvements in the overall sanitation of all aspects of dairy operations. Despite these well-intended efforts, episodes of milkborne illness still occurred in some AAMMC-certified dairies, including tuberculosis in the herd on the original dairy established by Dr. Coit himself (61).

From 1880 to 1915, following acceptance of the germ theory of disease and pasteurization, scientific interest in milk hygiene continued to grow as progress was made in improving the safety of milk and other dairy products relevant to human disease. This progress was evident from and attributable to an acceleration in the number of such publications in the medical literature (Fig. 2). Despite this progress, opposition to pasteurization increased in some areas because it was viewed as leading to unnecessary government regulation. For example, during the final quarter of the 19th century through the first quarter of the 20th century, the United Kingdom lacked consensus and public support regarding the need for regulating dairy farms and milk processing plants (2). It was not until 1922 that the United Kingdom required milk pasteurization (24). In contrast, as early as 1915 in major U.S. cities, most cow's milk was pasteurized; by 1936, 98% was pasteurized (46). In the United Kingdom, Atkins (2) reported that “only 1.5% of Britain's [dairy milk] supply was pasteurized in 1926. The majority of retail milk was still raw in 1939, and this remained true in many small towns and rural areas well into the 1950s.” This low rate of pasteurization occurred despite publication on both sides of the Atlantic of authoritative

texts with strong evidence supporting pasteurization (2, 53, 69).

FAVORABLE IMPACTS OF BREASTFEEDING AND ADVANCES IN FOOD SCIENCE ON INFANT MORTALITY

During the 19th century Industrial Revolution, infant mortality escalated to very high levels, especially in urban areas, and diarrheal disease was the most frequent cause of death. In London, for the 50-year interval from 1850 to 1900, epidemic diarrhea deaths among infants younger than 1 year of age increased by 41%: from 107 to 151 per 1,000 live births per year (43). Initially, the cause for this increase was thought to be substandard housing and poorly educated parents (40). However, from 1890 to 1910, poor food quality was recognized as an important factor (53). This phenomenon was referred to in Europe as the milk problem and in the United States as the milk question. During the 1880s, following the discoveries of Koch and Pasteur regarding the contribution of microorganisms to disease, contaminated milk was identified as a major vehicle for transmitting childhood illness, particularly summer diarrhea that occurred during the warmer months when bacteria multiplied more rapidly without refrigeration, resulting in an increase in infant deaths. Public health officials became well aware that this was a problem with some—but not all—solutions within economic reach, e.g., decrepit tenement housing, ignorant immigrant parents, and overall poor diet.

Between 1890 and 1910, with improvements in dairy hygiene, medical advances in infectious disease understanding, and expanding use of pasteurization came reductions in infant mortality to 80 to 180 per 1,000 live births per year, i.e., approximately half the rate of the mid-19th century but still not down to desirable or achievable levels. During this

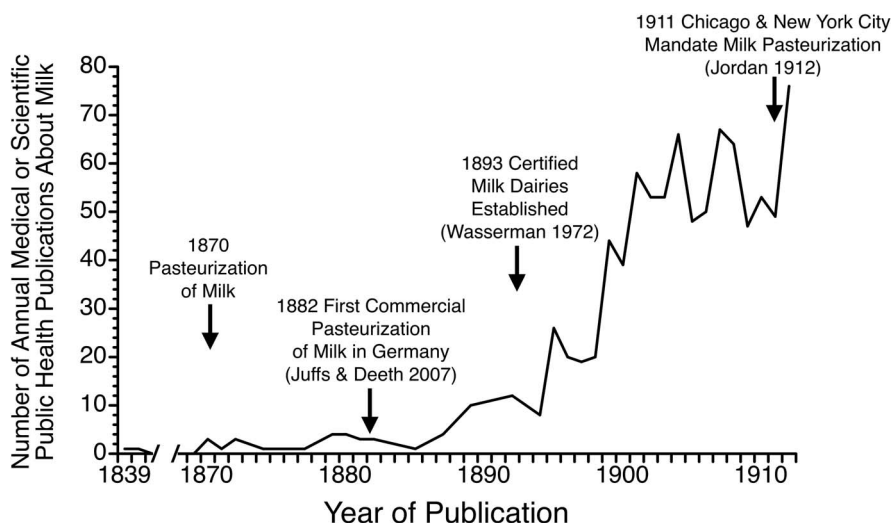


FIGURE 2. Increasing number of annual scientific publications about milk from 1839 to 1912, as modified from North (44).

20-year period, a number of American, French, German, and English studies revealed that only about 10% of infant deaths were associated with breastfeeding relative to the significantly higher percentage of deaths for nonbreastfed infants. In one study in NYC, of a “total of 1,065 infant deaths from diarrheal diseases, 16.7% of the babies were breast fed exclusively, 22.7% were fed on cow’s milk and breast milk combined, while 60.6% were exclusively artificially fed” (3). In another example reported by Meckel (40), “Investigating the circumstances in which 6,000 infants died in Liverpool [England], E.W. Hope concluded that artificially fed babies less than three months of age were 15 times more likely to die than breastfed babies” (47). In 1890, Pierre Budin, a French obstetrician, and W. J. Maloney (7) in a study of Paris infants younger than 1 year of age found a similar protective effect of breastfeeding (Fig. 3).

Additional scientific advancement occurred in the commercial production and availability of infant foods.

Food scientists discovered that cow’s milk could be modified to be similar to breast milk in its three major components: carbohydrate, fat, and protein. At the time, commercially produced infant foods contained a variety of ingredients, including farinaceous components (e.g., flour and meal), sugar (or molasses), and evaporated cow’s milk. Brand names for these products included Ridge’s Food from England, Nestlé Milk Food, Mellin’s Food, and Horlick’s Malted Milk (1, 26, 40). As the availability of improved commercial infant formulas increased, interest in breastfeeding declined. Although largely unrecognized, this movement away from breastfeeding occurred primarily among mothers forced to work out of the home and was accelerated by the invention of the India rubber bottle nipples (8).

ROLE OF HUMAN MICROORGANISMS IN SPECIFIC MILKBORNE DISEASES

Among early milkborne infectious outbreaks, the most frequent contributor in both the Europe and the United

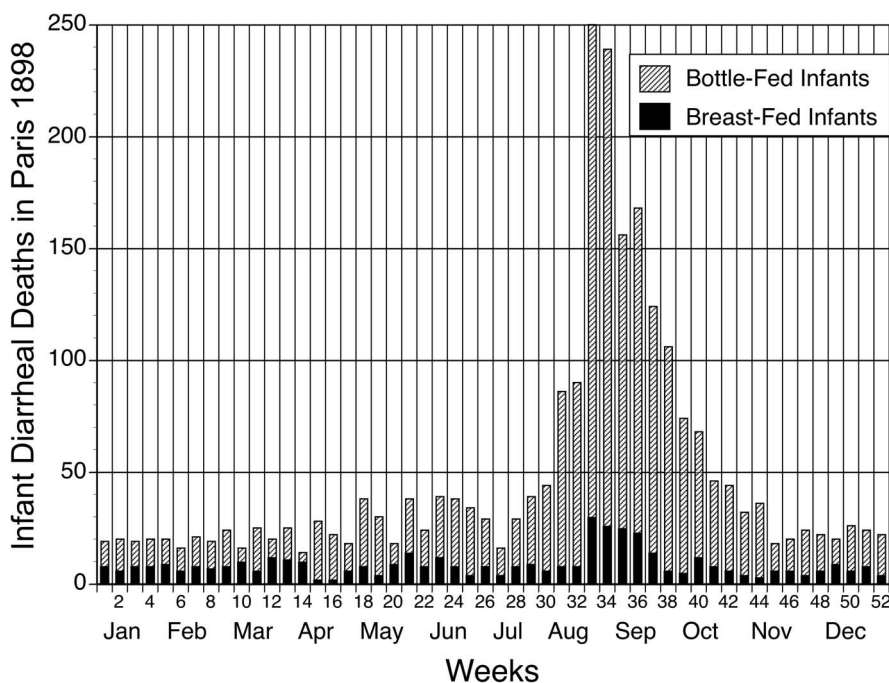


FIGURE 3. Weekly infant deaths among bottle-fed (shaded bars) and breast-fed (solid bars) infants in Paris, France, in 1898. Modified from Budin and Maloney (7) written when infant mortality in Paris was 288 deaths per 1,000 live births: “Favor breastfeeding by all possible means and provide mothers with advice and help in order to provide the best nutrition for the infant.”

States was typhoid fever, caused by *Salmonella* Typhi and *Salmonella* Paratyphi. One particularly important typhoid fever outbreak near Königsberg, Germany, was traced to a woman who had had typhoid fever 17 years earlier but who had remained both stool and urine positive for the pathogen (53). Thus, the lifelong typhoid carrier state of some individuals became recognized. A milkborne typhoid fever epidemic in Washington, D.C., prompted President Theodore Roosevelt to order a study of milk hygiene in 1905. Although the Pure Food and Drug Act of 1906 that Roosevelt also commissioned provided needed additional federal regulatory support for all dairy milk products (65), the landmark 1909 Rosenau report provided the unequivocal recommendation that all milk be pasteurized (52). Although in 1910 Chicago had mandated that all retail milk be pasteurized, effective implementation of this ordinance required several more years, largely because of technical problems with the flash pasteurization method (18).

Streptococcal infections were the second most common milkborne diseases of known etiology, and most cases resulted from human transmission. The Rosenau report of milkborne bacterial outbreaks included a table detailing 51 U.S. and British "scarlet fever" outbreaks caused by Group A streptococci from 1887 to 1907 (52). In 1911, a major epidemic of over 10,000 cases of streptococcal sore throat, including 19 fatalities, occurred in Chicago. The outbreak was linked to a large dairy plant still using a flash pasteurization apparatus but with inconsistent temperature control (9). Subsequently, the batch or vat method was developed, wherein milk was held for a longer period but at a lower temperature, most commonly at 140°F for 20 min (53). This batch method was used until after World War II, when processing at high temperature for short periods became the standard.

Diphtheria, caused by *Corynebacterium diphtheriae*, was the third most prevalent serious milkborne human disease. A table prepared by Hart in 1897 and published in the Rosenau report (52) included 146 diphtheria deaths among children and adults in Wales, Scotland, and England. The proven and suspected sources for these infections were highly variable and included farm-related contamination from hand-milking cows, contamination from milking utensils and bulk dispensing equipment in retail and home delivery settings, and transient infection of cows' mammary glands.

ROLE OF BOVINE MICROORGANISMS IN MILKBORNE DISEASE

Two additional important milkborne diseases, tuberculosis (both bovine and human) and brucellosis, can be transmitted to humans via raw milk. In 1896, Theobald Smith (57), a physician employed by the U.S. Department of Agriculture Bureau of Animal Industry, successfully demonstrated that bovine tuberculosis organism, *Mycobacterium bovis*, was distinctly different from the most common human tuberculosis organism, *Mycobacterium tuberculosis*. This difference was verified microscopically and by culture characteristics, laboratory virulence in animals, and the ability to produce more extensive lesions in cattle. Smith was also able to isolate the bovine bacillus from the

mesenteric lymph nodes of a child, thus fulfilling Koch's postulates (44, 54).

Although *M. bovis* is not as common a pathogen in humans as is the human *M. tuberculosis*, it is particularly invasive among children (54). The common portal of entry of *M. bovis* in children is the gastrointestinal tract; infrequently it enters through the lungs. As a consequence, presenting signs of human *M. bovis* infection frequently include one or more extrapulmonary sites such as abdominal lymphadenitis (85%), cervical lymphadenitis (50%), tuberculous lupus (49%), tubercular meningitis (25%), and bone and joint tuberculosis (20%) (20, 22). The estimated prevalence of human *M. bovis* infection in 1900 was 5 to 10% of all tuberculosis cases, with some locales reporting higher rates during the prepasteurization era (63, 64).

Between 1890 and 1910, a major problem in assessing the impact of *M. bovis* infection in humans was Koch's incorrect view that only one unique species-specific pathogenic mycobacterial species infected cattle. Because of growing scientific evidence at the time on both sides of the Atlantic that bovine tuberculosis was indeed a serious milkborne human illness, there was strong opposition to Koch's judgment. To settle this important issue, the German and British governments each appointed their own study commissions. In 1907, the Germans reported that *M. bovis* did indeed infect humans. Likewise in 1911, the British Commission reported that although *M. tuberculosis* caused the vast majority of cases of tuberculosis in adults and children, *M. bovis* also infected humans (54, 62).

The second most important bovine transmissible human disease of cattle was contagious abortion, known today as brucellosis (4). Initially the disease was referred to as Bang's disease, after its Danish discoverer, Bernhard Bang, who in 1897 identified the *Brucella abortus* organism in cattle and in raw milk. In the 1920s following recognition of infections in humans, brucellosis was determined to be widespread in both Europe and the United States. In 1931, Meyer (41) reported the presence of *Brucella* in AAMMC-certified milk. Although infrequently fatal, brucellosis has an acute, insidious onset, with a large constellation of symptoms including intermittent fever, headache, weakness, profuse sweating, arthralgia, myalgia, fatigue, anorexia, and weight loss.

Brucellosis was first identified by Bruce and Zammit on the island of Malta as part of the Mediterranean Fever Commission investigation of illness among military personnel (16). In reports between 1904 and 1907, Bruce used an agglutination test to study island goats, confirming infection. He reported that "Much to the surprise of the commission, several of them [goats] were found to react naturally to the agglutination test, and this led to the discovery of the *Micrococcus melitensis* [later *B. melitensis*] in the blood, urine, and milk. Several thousand goats on Malta were then examined, resulting in the epoch-making discovery that half of the goats on Malta responded to the agglutination test, and that 10% of the goats were secreting micrococci in their milk. Monkeys fed milk from an affected goat, even for one day, almost invariably took on the disease" (16). In response to these findings, requisition and consumption of goat's milk

in Malta was banned by British troops, and Malta fever ceased to exist among garrisoned troops.

In 1917, a distinguished American microbiologist, Alice Evans, conducted groundbreaking brucellosis research, demonstrating for the first time the close linkage between Malta fever in goats and epizootic abortion in cattle and infectivity in humans through infected cow's milk (58). As a result, Evans became an enthusiastic proponent of pasteurization (12), although as a consequence of her laboratory work, Evans herself later became infected with *B. melitensis* and suffered from the disease for over 20 years.

NONMILKBORNE FACTORS, SOCIAL PROGRAMS, AND TECHNOLOGY IMPACTING INFANT MORTALITY

Although the evolution of effective milk pasteurization from 1875 to 1925 did not totally resolve prevalent infant disease problems and high infant mortality rates (Fig. 1), it did accelerate the declining rates of milkborne infectious diseases in infants. However, a number of other factors contributed significantly to the reduction in infant infectious disease and resultant mortality.

Education was one such factor. In both Europe and the United States, families received education in basic hygiene measures. One example from 19th century Sweden was a national cadre of licensed nurse-midwives established to attend births and to educate mothers regarding proper infant care and feeding. When applied in conjunction with pasteurization, this approach led to a dramatic decline in Swedish infant mortality (6). In the United States, pediatrician S. Josephine Baker established impactful educational urban milk stations where nurses and other staff educated mothers about breastfeeding, home hygiene, and infant care. In 1907, Baker established the NYC Bureau of Child Hygiene and became its first director. An important facet of the work carried out at the NYC Bureau milk stations was the education of older children in the family who were deemed capable of assisting their mothers (3). Although not associated with the NYC Bureau, Lillian Wald, a medical student, dropped out of school to address the overwhelming need for educating NYC tenement mothers in proper infant care and feeding practice. Thus, Wald became the first public health nurse (a term she coined) while establishing the nation's first visiting nurse program (21).

These educational programs had a powerful impact on reducing infant mortality through the establishment of thorough, serial oversight home visits, medical referrals, and improvements in sanitation—including rat control. In her review of pediatric medical care up to 2012, Dr. Margaret Hostetter (27) noted that “as the 20th century dawned, infant deaths due to summer gastroenteritis failed to decline, despite mandatory pasteurization and the best efforts of the milk stations. Over the 10 year period from 1907 to 1917, the infant mortality rate decreased from 144.4 to 88.8 deaths per 1000, and the continued implementation of practical principles of household hygiene and child protection . . . led to a dramatic fall in childhood deaths from diphtheria 10 years before the vaccine was in widespread use.”

The role of education as a factor in reducing infant mortality was taken in a different direction by Sir George Newman (43), a British physician and pioneer in the fields of public and child health, who observed that infant mortality in England varied geographically and by social class. He reported that large towns had a rate of 172 deaths per 1,000 live births, whereas rural towns in England and Wales had a rate of 138 deaths per 1,000 live births. With regard to social class, Newman cited an 1898 study in York, England, in which infant deaths were categorized by class: the “poorest working class,” “middle working class,” and “highest working class” had infant mortality rates of 247, 184, and 173 per 1,000 live births, respectively. In general, rural versus urban rates had reversed themselves by the early 20th century. Improved dairy hygiene and pasteurization in cities likely played roles in the reversal of these trends (70).

The beginning of the second decade of the 20th century was a turning point for improvements in the quality of cow's milk formula and in a decline in infant mortality. As noted by Rollet and Bourdelais (51), “in the practical demonstrations of the obstetrician Pierre Budin, public action took various directions, including controls of the bacteriological quality, conservation and distribution of milk.” These measures included pasteurization and distribution through *goutte de lait* (“drop of milk”) centers. These centers also provided medical consultations for mothers about feeding, weaning, and appropriate breast milk substitutes.

In other European nations, epidemiological data gathered from various culturally distinct groups also informed and impacted infant health care, breastfeeding, and infant disease and mortality. These data included the influences of a nomadic existence among northern Scandinavian peoples with their higher summertime infant mortality and infant mortality rates that were lower in Austrian alpine environments than in nonalpine regions, lower in Finnish farm households than in nonfarm households, and higher in Swedish seasonal fishing areas than in nonfishing areas. These data clearly indicated that the influences on and trends in infant mortality within and among nations were complex and multifactorial (14). Newman (43) summarized as follows: “The causes of infant mortality are composite. It has been well said that every effect has an ancestry of causes. Pre-eminently this is the case in regard to infant mortality, which is due to a combination of factors closely inter-related.” Although the pasteurization of milk played a significant role, it was not the sole commanding influence leading to improvements in infant mortality.

From an economic perspective, the role of women in the workplace has been a factor that has impacted infant health and well-being. Although workplace conditions do not exclude breastfeeding, they can present a barrier, posing particular problems for women who are the primary or sole breadwinner. Boostrom (5) noted that “we human beings have never resolved the contradictions embedded in common notions about breast-feeding. We cannot decide if it is exaltation or tedium, fulfillment or drudgery, love or defilement.” Social and biological dimensions of breastfeeding have impacted infant mortality throughout human history. Over the 50-year period included in this review,

the merits of breastfeeding were once again confirmed during a period when cow's milk was well documented to be an infectious disease risk and nutritionally inferior.

In underdeveloped societies, the minimum nursing period is about 9 months, an age after which infants first begin to experience independence. Although breastfeeding of term infants has been repeatedly shown to be of great value in modern society, the provision of expressed human breast milk can be particularly beneficial in special feeding situations such as those of premature and distressed infants cared for in specialized neonatology centers. Although breastfeeding is a nutritionally sound, natural practice, milk banks accredited by the Human Milk Banking Association of North America pasteurize this product as a final safeguard (56).

UNSUNG HEROES THAT HAVE FAVORABLY IMPACTED MILK HYGIENE AND INFANT MORTALITY

Three additional groups of unsung heroes should be mentioned in this historical review of milk hygiene and its impact on infant mortality. The first group is the mid-19th century pediatricians, obstetricians, and medical health officers who recognized the risks of cow's milk for infant health. These thoughtful, visionary pioneers were shunned and belittled by mainstream medicine for their unorthodox views that subsequently were proven correct.

The second group is the visiting public health nurses, midwives, and doulas from many nations who helped educate mothers on hygienic infant care and the benefits of breastfeeding. These groups in turn motivated their peers in the Women's Suffrage movement from 1900 to 1920, prompting the U.S. Congress to fund improvements in public health in their passage of the Promotion of the Welfare and Hygiene of Maternity and Infancy Act, i.e., the Sheppard-Towner Act in 1921 (42). The effects that these groups had on decreasing infant mortality became obvious as the previously high breastfeeding rates were restored.

The third group of unsung heroes is the developers of the first effective pulsating vacuum milking machine for cows, which came into use in the United States in 1923 after a quarter century of development. The equipment closely imitated the sucking action of a nursing calf. Earlier milking machines had developed a bad reputation for ruining good cows (25). The new milking machines provided a clean, closed environment for cow's milk to be drawn away from dust, dirt, flies, cow feces, and other debris commonly found in milking sheds and barns. Most histories of dairying fail to mention this important development (39, 55, 66).

SUMMARY AND IMPLICATIONS FOR TODAY

The 50 years from 1875 to 1925 was a period of turbulent, complex, and continuous rapid change for Western nations. Changes, both favorable and unfavorable, impacting infant mortality occurred largely as a result of the Industrial Revolution and the many downstream effects it set into motion. Major changes included urbanization with resultant larger cities, a shift from an agrarian to an urban lifestyle, unregulated and often unscrupulous dairy business

practices, and the greater need for and reliance on scientific evidence. The Industrial Revolution significantly impacted the milk and dairy industries through its effect on milk hygiene and consequently on infant mortality. Just as in the past, today's postneonatal infant mortality rate is highly sensitive to environmental influences and functions as society's environmental barometer of overall health and well-being, i.e., the "canary in the coal mine." As illustrated in this review, the many concurrently evolving practices and events taking place in various geographic locations around the world from 1875 to 1925 make it difficult to precisely quantitate the impact of the many complex, sometimes interrelated factors. Infant mortality in the United States was clearly impacted by the distillery industry with its close association with the dairy industry for increasing profitability (Fig. 1) (35). During this period, important public health practices were discovered or reaffirmed, including the protective effects of breast milk compared with cow's milk for feeding infants, the adverse effects of urbanization, and the enlightening effects of advances in scientific inquiry, beginning with proof of the germ theory by Pasteur and Koch. Although current infant mortality rates in developed countries have declined to 5 to 6 deaths per 1,000 live births per year, the germ theory continues to have profound implications and effects, resulting in continued improvements in milk hygiene and declining infant infectious disease and mortality. Unfortunately, some well-recognized preventable contributors to infant disease and mortality persist to the present; among these contributors is the effort to commercialize the sale of raw milk (15). Only when we are able to learn from the scientifically verifiable mistakes of the past with respect to milk hygiene and the importance of breastfeeding will we be better able to guarantee public health improvements for the future.

EPILOGUE

In 1940, the prevalence of breastfeeding in the United States was 65%. By 1958, it had declined to 25%, where it remained until the early 1970s (19). Subsequently, breastfeeding for the first 6 months of age gradually rose to its present prevalence of 51% (10). Recent breastfeeding prevalence in 153 countries by income levels of low, lower-middle, upper-middle, and high income was 98, 97, 75, and 45%, respectively (67). Although advice from the World Health Organization, UNICEF, and other respected professional health organizations is that "breast is best," a contemporary dilemma presents itself for mothers who are HIV positive (13) or are infected with cytomegalovirus (50) and thus are discouraged from breastfeeding.

Discussions of the status of cow's milk as an acceptable alternative nutrient source for infants and of the benefits of pasteurization of milk continue to the present. Pasteurization technology required 50 years of incremental change from 1900 to 1950 to achieve acceptability. The last U.S. AAMMC-certified dairy closed in California in 1999 after merging with a national dairy company. Up to the time of closure, consumption of this operation's raw cow's milk carried increased risks of infection, e.g., with *Salmonella enterica* serotype Dublin, an invasive pathogen in children (17).

Although still consumed today, raw milk continues to pose risks from a disease standpoint. Numerous reports in the United States have revealed disease risks associated with consumption of raw milk, including risks of salmonellosis, campylobacteriosis, and hemorrhagic *Escherichia coli* O157:H7 infection (36, 37, 45, 48). Currently in the United States, 1 to 3% of all milk sales are of raw milk. The sale of raw milk is permitted under a variety of legal venues in 40 of the 50 states, but interstate sale is not permitted. In the absence of structured oversight by the AAMMC, raw milk dairies reflect a range of quality from fairly good to marginal, posing considerable disease risks (30). In a recent report, consumption of unpasteurized dairy products was associated with 840 times more illnesses (95% credibility interval, 611 to 1,158 times; the Bayesian equivalent of the confidence interval) and 45 times more hospitalizations (95% credibility interval, 34 to 59 times) than was consumption of pasteurized dairy products (15). Critical reviews examining the benefits of raw cow's milk over pasteurized cow's milk have failed to reveal evidence of any nutritional, antimicrobial, intrinsic enzyme, allergic propensity, immunity, or other disease advantages of one type of milk over the other (11, 38). With no benefit and the high risk of infectious disease accompanying raw milk consumption, pasteurization of milk is overwhelmingly preferred.

Both sides of breast versus bottle feeding and raw versus pasteurized milk consumption continue to have their advocates, and these debates seem certain to continue well into the 21st century. Scientists, public health officials, food safety professionals, health care workers, and consumers must choose among the various types of evidence based on their personal knowledge, culture, geography, and other influences. As expressed two millennia ago by Cicero in his *De Finibus Bonorum et Malorum* (I, 13): “Through ignorance of what is good and bad, the life of men is greatly perplexed.”

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